

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics
Course Code	STA2MN102 (P)
Course Title	Probability theory II
Type of Course	Minor

Semester	II				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic Knowledge in the concept of Probability and Random Variables				
Course Summary	Students will possess a comprehensive understanding of bivariate random variables, enabling them to analyze and interpret the joint behavior of two random variables.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define and explain the concept of bivariate random variables.	U	C	Instructor-created exams / Quiz
CO2	Explore the concept of joint and marginal probability density functions	U	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Calculate bivariate expectations for various functions of two random variables	Ap	F	Seminar Presentation / Group Tutorial Work/Instructor-created exams
CO4	Identify and describe common standard distributions like rectangular, beta, gamma and analyze data to help entrepreneurial decisions using critical thinking skills.	U	C	Instructor-created exams / Home Assignments
CO5	Identify the importance of time series analysis and explain various components of time series and critically evaluate ethical implications of statistical methods aligning with human values.	U	F	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Locate probability curves for different distributions using R	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

COURSE CONTENT

Module	Units	Content	Hrs (45 +30)	Marks (70)
I	TWO DIMENSIONAL RANDOM VARIABLES		11	20
	1	Introduction to two dimensional random variables	1	
	2	Joint PMF and Joint pdf (Concept and Problems)	2	
	3	Joint DF(Concept and Problems)	2	
	4	Marginal Distributions(Concept and Problems)	2	
	5	Conditional Distributions(Concept and Problems)	3	
	6	Independence of Random Variables(Concept and Problems)	1	
	Sections from References: Unit 1: 5.5 [Ref 1] Unit 2: 5.5.1 & 5.5.4 [Ref 1] Unit 3: 5.5.2 [Ref 1] Unit 4: 5.5.1 & 5.5.4[Ref 1] Unit 5:5.5.1 & 5.5.5 [Ref 1] Unit 6:5.5.6 [Ref 1]			
II	BIVARIATE EXPECTATION		12	15
	7	Expectation of two random variables (Concept and Problems),Addition Theorem (Statement Only), Multiplication Theorem (Statement Only)	3	
	8	Properties of Variance	1	
	9	Covariance & Correlation Coefficient	3	
	10	Conditional Expectation and Conditional Variance (Concept and Problems)	5	
	Sections from References: Unit 7: 6.4 [Ref 1] Unit 8: 6.5 [Ref 1] Unit 9: 6.6&10.4 [Ref 1] Unit 10: 6.9[Ref 1]			
III	STANDARD DISTRIBUTIONS		12	15
	11	Discrete Uniform Distribution (Mean, variance, mgf,	1	

		Problems)		
	12	Geometric Distribution (Mean, variance, mgf, Problems)	1	
	13	Hypergeometric Distribution (Mean, variance, mgf, Problems)	1	
	14	Negative Binomial Distribution (Mean, variance, mgf, Problems)	1	
	15	Rectangular Distribution(Mean, variance, mgf, Problems)	2	
	16	Gamma Distribution(Mean, variance, mgf, Problems)	2	
	17	Beta Distribution(Mean, variance, mgf, Problems)	2	
	18	Order Statistics[Distribution function of single order statistic ,Examples]	2	
	Sections from References: Unit 11: 8.2 [Ref 1] Unit 12: 8.7 [Ref 1] Unit 13: 8.8 [Ref 1] Unit 14: 8.6[Ref 1] Unit 15: 9.3[Ref 1] Unit 16:.9.5 [Ref 1] Unit 17:.9.6,9.7 [Ref 1] Unit 18:.9.18 [Ref 1]			
IV		TIME SERIES ANALYSIS	10	20
	19	Introduction to Time Series & Utility of Time Series	1	
	20	Components of Time Series	1	
	21	Measurment of Trend- Graphic Method,Semi Average Method, Method of Moving Average, Method of Least squares (Linear Trend) (Concept and Problems)	4	
	22	Measurement of Seasonal Variations-Method of Simple Averages ,Ratio to Trend Method	4	
	Sections from References: Unit 19: 11.1& 11.3[Ref 2] Unit 20:11.2[Ref 2] Unit 21:.11.5[Ref 2] Unit 22:.11.6 [Ref 2]			
V	PRACTICUM		30	
	Do practice problems in R software from any 5 units of the given list			

	and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Measures of skewness		
	2	Measures of kurtosis		
	3	Obtain the probability distribution		
	4	Plot the probability distribution		
	5	Obtain the cumulative distribution function		
	6	Plot the cumulative distribution function		
	7	Obtain any one discrete probability		
	8	Obtain any one continuous probability		
	Sections from References: Unit 1: 2.6 [Ref 5] Unit 2: 2.6 [Ref 5] Unit 3: 3.2, 3.3 [Ref 5] Unit 4: 3.2, 3.3 [Ref 5] Unit 5: 3.2, 3.3 [Ref 5] Unit 6: 3.2, 3.3 [Ref 5] Unit 7: 3.4 [Ref 5] Unit 8: 3.6 [Ref 5]			
Books and References: 1. Gupta, S. C. and Kapoor, V. K. (2020). Fundamentals of Mathematical Statistics, 12 th edition, Sulthan Chand, New Delhi 2. Gupta, S. C.. (2015). Fundamentals of Statistics. , & 7 th edition, Himalaya Publishing House 3. Gupta S.C (2021), Statistical Methods, 46 th edition, Sultana Chand and Sons. 4. The R book(2007) , Michael J. Crawley John Wiley Series 5. Sudha G Purohith, Sharad D Core, Shailaja R Deshmukh ,Statistics Using R(2015)				

Mapping of COs with PSOs and POs :

	PSO	PSO	PSO	PSO4	PSO	PSO6	PO	PO2	PO3	PO4	PO5	PO6
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	1	2	3		5		1					
CO 1	2	3	-	-	-	3	1	2	2	-	-	-
CO 2	-	2	-	-	-	-	2	-	-	-	-	-
CO 3	-	3	-	2	2	2	1	2	-	-	-	-
CO 4	-	-	3	-	-	3	3	-	1	3	3	-
CO 5	3	-	-	-	-	2	3	1	-	-	-	3
CO 6	-	-	-	-	-	2	2	3	-	-	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4		✓		✓
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CO 6	✓			